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RESEARCH ARTICLE

Deep Learning-Based Assistive Robotic Grasping System for Visually Impaired Individuals

MUHAMMAD ARSALAN¹, ABID IMRAN¹, ARSALAN ARIF¹, KHURRAM KHAN²,
AND MUDASSIR RAZA SIDDIQI³

¹Faculty of Mechanical Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Swabi, Topi 23640, Pakistan

²Faculty of Computer Science and Computer Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Swabi, Topi 23640, Pakistan

³Department of Electrical Engineering, KTH Royal Institute of Technology, 10044 Stockholm, Sweden

Corresponding author: Mudassir Raza Siddiqi (mrsi@kth.se)

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ABSTRACT Robotic assistive systems play an important role in enhancing quality of life of visually impaired individuals with greater independence. A significant amount of research has been conducted in assistive grasping systems in recent years, but certain challenges still exist including background noise during voice-based interaction, object occlusion, reliable and safe object handover in human robot interaction (HRI), and multimodal interfaces. To address these challenges, this research presents an intelligent assistive robotic grasping framework that integrates the vision module, voice module and robot control module. The vision module includes the enhanced YOLOv11s-seg model that performs unified object segmentation with pose estimation and achieved mAP@50-95 of 0.933 for detection and 0.882 for segmentation with the least computational cost (32.8 GFLOPs) and fast inference time (4.2 ms/image) compared to YOLOv8s and YOLOv9c. The voice module includes a hybrid band-pass with spectral filter that efficiently removes the background noise (low, medium and higher background noise) to convert voice command into texts with an accuracy of 93.5%. Additionally, an SVM-based machine learning text classification system is employed to enhance safety by rejecting voice commands unrelated to grasping or object picking, achieving 97.93% accuracy. Finally, the robot control module is introduced with object grasping and safe handover algorithms. The proposed intelligent assistive robotic grasping framework is validated through different experiments on sighted users which demonstrate the overall average grasp success rate of 94.7%, with a standard deviation of $\pm 7.76\%$ across thirteen daily-life objects, indicating variability in performance due to object specific challenges. The system demonstrates an average execution time of 42.67 ± 1.47 s from the voice command to the system to handing over the grasp object to user by robotic arm. The system ensures operational safety through workspace boundary constraints and built-in force sensing. The proposed approach provides a novel, efficient, and human-centric solution for voice-guided assistive robotic grasping for visually impaired users.

INDEX TERMS Assistive grasping, deep learning, segmentation, voice recognition, visually impaired, you only look once (YOLO).

I. INTRODUCTION

A significant portion of our society consists of people with different types of disabilities, either inborn, diseases, or after accidents [1]. These individuals may face difficulty performing daily life activities, i.e., walking, grasping, sensing,

seeing, etc. According to a survey conducted in the US in 2015, the number of people above 40 years of age who suffer from blindness and vision impairment is 1.02 and 3.22 million respectively [2]. According to the NIH (National Institutes of Health), the number of blind or visually impaired people is expected to more than double by 2050, to approximately 8 million [3]. The WHO (World Health Organization) reported that after the COVID-19 pandemic, to minimize

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human contact, nearly 70% of countries affected provided rehabilitation services [4], [5]. In response to these growing needs, recent advancements in intelligent robotic platforms, sensors, control systems, and intelligent machine vision algorithms, robots have the capability to provide rehabilitation and assistance to people with different disabilities, improving their quality of life [5], [6]. Robotic manipulators are used in a variety of sectors ranging from education, automotive, manufacturing industries, social services and for rehabilitation and assistive purposes, where disabled persons need attention and support. These disabilities can be motor impairments, such as requiring prostheses or orthoses, and sensory impairments, such as visual or hearing disabilities. For both applications, the size of robotic arm varies from few inches to several feet [7]. More recently with the development of the robotic system they are mostly used for the biomedical sector, especially for rehabilitation and assistive purposes, which has helped individuals both socially and personally such as improving their quality of life, self-respect and independence [8], [9]. Despite advances in robotic systems, it requires further improvements in control, usability, and human–robot interaction (HRI) safety [10]. Traditional joystick based control techniques for assistive robotic application demands precise hand motion and multiple cognitive senses to visually see and control but this approach is challenging when the DOF (Degree Of Freedom) increases or the user with physical limitations [11]. To overcome these limitations, alternative control techniques, including voice commands, eye tracking, touch based, graphical user interfaces (GUIs), and hybrid techniques have been explored to improve usability and efficiency in assistive robotic systems [12], [13], [14].

In assistive robotic grasping system different machine learning models and algorithms have been implemented, like supervised, unsupervised and reinforcement learning (RL), to improve the adaptability and autonomous functions operations [15], [16]. Recent literatures have applied RL to robotic grasping systems, demonstrating robustness without modeling the object geometry [17]. However, real-time implementation of RL-based assistive systems remains the concern due to safety, reward design complexity, and the need for extensive training, especially in human robot interaction involving visually impaired users [18]. Although, deep reinforcement learning models have addressed adaptive grasping and post-grasp manipulation in dynamic environments in simulation and controlled settings, and for real-time applications for visually impaired users still face challenging for intense simulation training demands and safety considerations [19].

These existing grasping and manipulation systems have become more challenging for visually impaired users when performing object manipulation and daily-life tasks. As the existing control techniques require visual feedback and visual understanding. Moreover, the visually impaired also find it difficult to precisely locate, recognize, and pick the target object from random environment. For example, retrieving the water bottle from refrigerator without assistance

is very challenging and dangerous for visually impaired individuals.

Despite recent progress, several challenges remain insufficiently addressed in assistive robotic grasping system for visually impaired users. These challenges include the inaccurate performance of voice commands recognition in noisy background, partially object occlusion, unsafe object handover to visually impaired users, and difficulties in integrating vision, voice and robot control modules into a single system.

To address these limitations, this paper proposes an intelligent system that integrates a robotic platform, sensors, and advanced artificial intelligence algorithms, aims to enhance the overall quality of life for individuals with visual impairments. For system evaluation, a voice-enabled robotic arm is developed to identify and grasp the target objects in an open environment, enabling visually impaired users to perform daily object manipulation tasks using voice commands. The proposed intelligent assistive grasping system consists of the three core modules: a machine vision module responsible for object segmentation and localization, a voice model for command recognition, and a robot control module for safe and accurate manipulation. Each module is optimized to enhance overall performance. A significant achievement of this research lies in the smooth integration of these three modules into an intelligent assistive grasping system designed for visually impaired individuals. The effectiveness of the proposed system is validated through real-world experiments on sighted participants, and the results demonstrate its potential applicability for assisting visually impaired individuals in daily life activities. Moreover, the proposed framework is flexible and can be extended to support users with other types of disabilities. The main contributions of our research are:

- Proposed an algorithm that detects and segments objects and estimates the grasp pose for all objects in the workspace using an improved YOLOv11s-seg model. The algorithm also enables safe grasping of the targeted object in occluded scenarios without damaging non-target objects.
- A voice module, which can recognize and convert speech into text even in moderately noisy environments by applying a hybrid band-pass and spectral filter, along with an SVM-based text classification system (97.93% accuracy) to prevent unsafe or ambiguous command execution.
- Hand detection, tracking, and hand motion stability for efficient and safe handover of the object to the user.
- Seamless integration of three different modules (Machine Vision Module, Voice Module, and Robot Control Module) into a single intelligent assistive grasping system.

The rest of the article is organized as follows: Section II is a review of the study conducted on assistive robotic grasping; Section III is about methodology; Section IV explains the results, Section V presents the overall system latency and performance analysis, Section VI presents the failure cases

TABLE 1. Comparison of existing assistive grasping system for visually impaired individuals.

Ref.	Approach / System Description	Key Technologies Used	Vision Model	Advantages / Strengths	Limitations / Research Gaps
N. Nandan et. al [30]	Kinect-based personal assistant robot for visually impaired individuals; gesture & voice-controlled 6-DOF robotic arm.	Kinect vision sensor, microphone, 6DOF robotic arm.	Faster RCNN	Enables multi-modal interaction (voice + gesture); assists with daily tasks.	- No safe handover system. - limited precision in grasping.
J. G. Victores et. al [31]	Voice and 3D-vision-controlled robot for users with disabilities.	Voice recognition module, stereo vision system.	Color segmentation/ No object detection	Enables voice-controlled motion without joystick; suitable for limited mobility users.	- Voice-based control not optimized for visually impaired users - lacks autonomy and object-pose awareness.
H. W. Ka et al [33], T.B. Pulikottil et. al [34], and H. Ka et. al[35]	Semi-autonomous assistive robot using voice commands for arm movement and gripper actuation.	Computer vision + voice recognition modules, 6-DOF manipulator.	-----	Removes manual control dependency; supports hands-free operation.	- Tedious and sequential control(up/down/left/right). - slow and cognitively demanding - lacks automatic trajectory planning.
R. A John et. al [35]	Voice-controlled robotic arm for visually impaired persons.	Speech recognition + microphone input.	ImageNet	Natural interaction through speech.	- Poor robustness to ambient noise. - accuracy decreases with distance between user and microphone.
G. Nantzios et. al [36]	Niryo robot system integrating vision and voice sensors.	RGB camera, speech recognition.	YOLOv5	Provides visual assistance and remote operation.	- Struggles with background noise and pronunciation variations. - low command accuracy.
S. Rakhimkul et. al [72]	Deep learning based assistive robotic grasping system	RGB-D camera, GUI based object selection	YOLOv3	Human intent prediction for user's joystick control action.	- Target object occlusion. - Poor robot motion planning with joystick.
T. Shahria et. al [73]	Activities of Daily Living Assistance Using Assistive Robot	RGBD camera, GUI based object selection	YOLOv5s	Vision based grasping system but tested on symmetric round shapes	- Always side wise grasping. - Not included orientation control - top-down grasping not possible.

and future work, and Section VII presents the conclusion of the proposed system.

II. RELATED WORK

A. ASSISTIVE ROBOTS

Table 1 summarizes the existing assistive grasping systems developed for visually impaired users. Existing approaches/work integrates computer vision and voice-based control framework using depth cameras, microphones and speech recognition modules. While the systems proved reliable with gesture and voice interaction in controlled environment, performance is degraded for precise grasp planning in occlusion, sequential control, and no autonomous trajectory planning. With the advancement in deep learning models such as YOLOv3 and YOLOv5 detection/ segmentation predictions for recognizing objects has been improved significantly. However, the challenges in speech recognition under noisy environments, poor motion planning, an unfeasible handover mechanism and limited occluded scenarios handling still need to be addressed.

Research on assistive technology for visually impaired individuals has primarily emphasized enhancing

independence and safety in navigation within indoor and outdoor environments [20]. Smartphone-based setups with computer vision technology have been proposed to detect outdoor objects [21], while prototypes featuring smart sticks equipped with cameras and cloud APIs aim to aid in obstacle detection and avoidance [22], [23]. Though, besides navigation, robotic system can be used in variety of task for assisting visually impaired in their daily life activities, e.g. drinking, hair combing, medical assistance, assistive feeding, household assistance and also in object manipulation [24], [25], [26], [27], [28], [29]. Yet relatively less research has been conducted for assistive grasping with robotic manipulators for visually impaired for object manipulation and grasping.

To enhance the assistive robotic system for visually impaired different approaches were proposed but the most suitable was to focus on multimodal integration using computer vision and speech-based control techniques. Initially, Kinect vision sensor based prototype was developed which was integrated with speech recognition using microphones, and a 6 DOF robotic arm and demonstrated the functionality of the system using gesture and voice based operation of visually impaired user [30]. But without positioning control

TABLE 2. Evaluation and comparison of two stage and single stage object detectors.

Model	Ref.	Type	Region Proposal / Anchor Strategy	Speed (FPS)-GPU	mAP@50(%)
R-CNN	[74]	Two-Stage	Selective Search	--	58.5 (VOC)
Fast R-CNN	[75]		Selective Search	0.5	70.0 (VOC)
Faster R-CNN	[76]		RPN anchors	7	73.2 (VOC) / 34.9 (COCO)
Mask R-CNN	[77]		RPN + RoI Align	5	80.3 (COCO)
SSD	[44]	Single-Stage	Anchor boxes per layer	46	74.3 (COCO)
YOLOv1	[78]		None (grid-based)	45	63.4 (COCO)
YOLOv2			Anchor-based	67	76.8 (COCO)
YOLOv3				35	55.3 (COCO)
YOLOv4				65	43.5 (COCO)
YOLOv5				101	50.7 (COCO)
YOLOv6				121	52.8 (COCO)
YOLOv7				114	56.8 (COCO)
YOLOv8				Anchor-free	82
YOLOv9			73		55.6 (COCO)
YOLOv10			120		54.4 (COCO)
YOLOv11			240		54.7 (COCO)

system, limited adaptability and structured environment made the system less user friendly and reliable. Later, another prototype combined depth camera with voice recognition improved the system efficiency but were not suitable for visually impaired. Because the actuation system contained manual trajectory planning [31]. Then, the researcher focused on semi-autonomous assistive robotic system which also included the computer vision and voice recognition capabilities to control the 6DOF robotic manipulators [32], [33], [34]. These improved systems have eliminated the need for manual control; however, they still have challenges for trajectory planning instead of joystick but with the discrete voice command (move up, down, right and left, gripper open/close). Which still has lower speed and automation. Mostly voice command based control system performance is reduced due to background noise, accent, and varying user to microphone distance, which effects overall system accuracy [35], [36], [37].

B. MODEL SELECTION

Robotic grasping needs two important parameters: object segmentation and grasp pose estimation (which includes the grasp position and orientation), to successfully pick the object without damaging the robot and environment. Naturally, the human always grasp the objects along the long axis for stability and safety [38], [39], and thus implemented the same approach for robotic grasping and manipulation. Conventional machine learning models, e.g., SVM, and random forest, have been used in early assistive robotic system for grasp pose estimation. These approaches were not suitable for real time implementation due to slower inference speed because of hand crafted features input to the model and had lower generalization capability. But with development of neural networks, object segmentation and grasp pose estimation precision and efficiency had been increased which enabled them to predict in real time in complex scenarios [40], [41].

But still there was a gap for improvement regarding accuracy and inference speed.

Different deep learning architectures and techniques were introduced which further enhance the performance of the deep learning models, i.e. two stage models and single stage models [42], [43]. Two-stage models like R-CNN, Fast R-CNN, and Faster R-CNN performed well for detection/segmentation accuracy. But the higher number of layers which increased the parameters affect computational power and inference speed. Therefore, to achieve both performances in optimized way the researcher developed single stage architecture such as SSD [44] and YOLO1 [45], which enhanced both detection or segmentation accuracy with the lower inference time. Further improvement on YOLO models was performed from version1 to version11 and improved both speed and accuracy through innovations like CSP backbones, PANet fusion, and anchor-free decoupled heads. The recent versions YOLOv11 integrates C2PSA and C3k2 modules, improved detection of small and occluded objects while achieving high inference speed. Therefore, modern YOLO models balanced accuracy, efficiency, with real-time performance, which is suitable for assistive grasping applications [46], [47]. A comparative summary of these models is presented in Table 2, where YOLOv11 model exhibits higher real-time performance as well with high detection accuracy, less parameters, and fast inference speed, making it the most suitable model for assistive grasping applications.

C. TARGETED OBJECT GRASPING

For robotic grasping, different models were developed and implemented from supervised, unsupervised and reinforcement learning (RL) [48], [49], [50], [51]. Each has its pros and cons regarding robotic grasping. Supervised learning trains on the labelled datasets and gives good accuracy and inference speed but it struggles on novel objects. Unsupervised and self-supervised model improves generalization

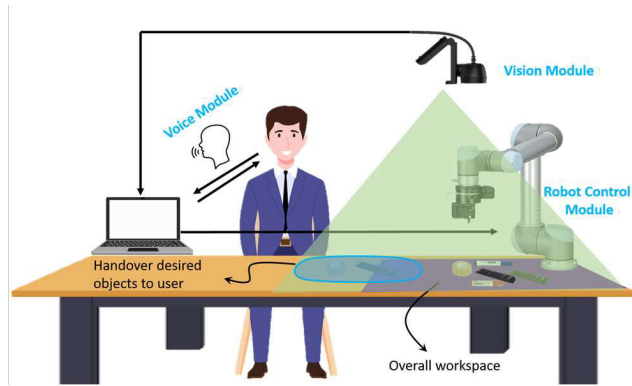


FIGURE 1. Overall intelligent assistive robotic grasping system for the visually impaired person. The overall system consists of a machine vision module, voice module, and robot control module.

by learning from sensor inputs directly, but struggles with precise action selection [48], [49]. RL based learning is achieved by autonomous grasp strategy learning through action and reward based optimization, performs well in cluttered, unstructured environment but requires reasonable simulation environment to train the model which needs high computational resources with higher training time [19], [52], [53]. Hybrid models are then the suitable choice to balance the accuracy, adaptability and inference speed, which showed reliable grasping performance in assistive robotic systems.

Hybrid models integrate object detection/segmentation models and grasp pose estimation models, e.g., combine YOLO for segmentation with CNN based grasp pose estimation model, enhance generalization, accuracy but with higher inference latency and computational cost. Robotic grasping is categorized into 6D and 4D pose estimation techniques. 6D pose estimation provides complete 6D object information, location and orientation in 3D space, therefore requires complex trajectory planning and training. On the other hand, 4D pose estimation provides top-down grasp configuration, where the orienting is only for the gripper finger orientation to adjust according to object shape. Therefore, reduces the computational power, training, and fast inference. Thus the 4D pose estimation technique is suitable for the real time assistive grasping system in moderately cluttered and structured tabletop scenarios [54], [55], [56], [57], [58], [59].

D. RESEARCH GAPS

From the literature review conducted for different assistive grasping with robotic arms highlights the promising advantages to improve the productivity and independence of visually impaired person. But still it needs to be improved to help the visually impaired to perform basic daily life activities like grasping/ pick and place with ease in usability, safety, and quick response. Some of the limitations are:

- 1) Existing literature used hybrid models (combines object detection model with grasp pose model), causing higher computational cost with slow response.

- 2) Background noise mostly hinders voice module efficiency. This noise interface with voice command of user cause wrongly recognizing the command.
- 3) Real scenarios often contain partial occlusion cases, which effects both detection and grasp planning.
- 4) Visually impaired users require adaptive handover methods; however, existing works partially address user-centric and real-time handover safety. A user centric solution is needed to handover object to user's hand to ensure safety and reliability of the system.
- 5) Seamless integration of vision, voice and robotic arm control modules is a major challenge for efficient and quick responsive assistive grasping system.

E. NOVELTY

The novelty of this work is summarized as:

1. Proposed a unified deep learning model-based system that detects, segments and 4D grasp pose estimation for target object.
2. Hybrid band pass with spectral filters technique is implemented to omit background noise, and to make sure only the user voice command is processed.
3. A novel grasp planning for partially occluded target object caused by non-target objects.
4. A vision based adaptive handover framework, which ensures the hand is detected, stable then perform the trajectory planning to handover. This shows the system adaptability and user centered interaction.
5. A seamless multimodal framework integration of vision, voice, and robotic manipulator control module, with low-latency response from voice command to object handover for efficient and real-time assistive grasping action.

III. METHODOLOGY

A. PROPOSED SYSTEM ARCHITECTURE

The development of an intelligent robotic manipulator system for visually impaired individuals, equipped with voice command functionality for assistive grasping, contributes to the field of assistive technology. This assistive robotic grasping system is designed to improve the autonomy and independence of visually impaired to easily perform their daily life activities, especially in grasping and object manipulations capability within the specified environment. The complete intelligent assistive grasping system using robotic manipulators is shown in Figure 1, which illustrates the overall integration of essential modules necessary for assistive grasping for visually impaired. The system architecture consists of three modules: The Machine Vision Module, the Voice Module, and the Robot Control Module. Each module plays a distinct role in ensuring the smooth operation of the entire system. The first step of the system involves scanning of the workspace by machine vision module, which is responsible to detect and recognize all the objects in the workspace to generate each object grasp poses and save this information of

objects placed in the workspace. This step lays the base for the subsequent actions of the robotic manipulator. Once the workspace scanning is completed, the system sends a voice message: “How may I help you?”. After that, the user can give two types of commands: one is to scan the environment or table to obtain visual information and announce the names of detected objects for the visually impaired, and the other is to command the system to pick or grasp a named object, which activates the robotic arm to pick it up and hand it over to the user. This is necessary for interaction between the environment and user to perform direct action instead to repeat the object names in every loop. The voice module helps to remove background noise, recognized voice, convert it into text and pass through the intelligent safety filter to check for grasping related voice command or irrelevant command. All these steps are important, as they help the system to understand the user’s command accurately. This intelligent safety filter increases the safety of the system to process and pass only grasping related commands to the robot control module for further action. This integration of machine learning enhances adaptability and ensures that only accurate instructions are transmitted to the robot control module for further action. This module helps to process the sequential robotic action movement to perform direct grasping or occlusion removal tasks. The seamless coordination among voice commands, machine vision, and robotic control showcases the system’s sophistication in meeting user needs. Once the task is successful for grasping the target object, the system is then again ready to receive another user voice command. This makes the system efficient, adaptive and user centered design.

B. MACHINE VISION MODULE

This module is assigned to generate dataset for segmentation model, which is then labelled and trained on YOLOv11s-seg model to detect, segment the objects, and extract the grasp poses of the objects. The machine vision module is further divided into sub-categories: data acquisition and the improved YOLOv11s-seg model for object segmentation with pose estimation.

1) DATA ACQUISITION

The dataset included thirteen different daily-life objects selected for assistive grasping applications for visually impaired patients. The objects are TV remote, medicine box, black markers, blue markers, juice, red markers, plaster paste, key, solder paste, bandages, medical taps, whitener, and AC remote. The original images in the dataset were 1985 images, and the randomness in the dataset was incorporated by capturing images at different positions, different orientations, randomly placed, and with varying lighting conditions in the laboratory. The images were captured using two different devices: A Rapoo C200 720 HD webcam, which recorded at a resolution of 1280×720 pixels at 30 fps, and a mobile phone equipped with a 48-megapixel RGB camera, which recorded at a resolution of 1920×1080 pixels. To enhance



FIGURE 2. Sample images of the dataset used during training. Thirteen distinct objects were placed on the table-top for dataset generation at different positions, orientations, different background tabletop, and different reflective lighting conditions.

computational efficiency, all images were uniformly resized to 640×640 pixels. The original images were labeled in Roboflow (an online annotation tool), which were then divided into training, validation, and testing sets at 80%, 10%, and 10% respectively. Then augmentation was applied only to the training set, resulting in a total of 5532 images. The respective datasets were created having 4426 images in the training set, 553 images in the validation set, and 553 images in the test set. We deliberately increased the difficulty of our dataset by including objects that share similar colors among themselves and with the background. To accommodate the diversity of the dataset generated, all the images of the dataset were captured under different variations factors like different colored backgrounds, objects in a cluttered environment, different variations of objects with other surrounding objects, under different lighting conditions (No reflective lights, medium reflective lights, and high reflective lights), objects in the middle of the table, and the objects at the corner of the table. When conventional image processing techniques are applied, conditions such as low color contrast between the target object and background and the different intensities of the reflective lights often lead to insufficient information extraction. For instance, the color of the remote in our dataset perfectly blends with the background, creating a challenging scenario. Similarly, markers predominantly have a whitish color, closely resembling the background color. Figure 2 shows sample images from our dataset to illustrate these challenging scenarios.

The object classes in the dataset are selected in a way that balances practicality and system complexity. Some classes consist of essential items that are important for visually impaired people in their daily lives, such as medicines, different pastes and bandages etc. In addition, several other objects are intentionally included to increase the diversity and complexity of the dataset. These additional objects may not be of critical importance to visually impaired people

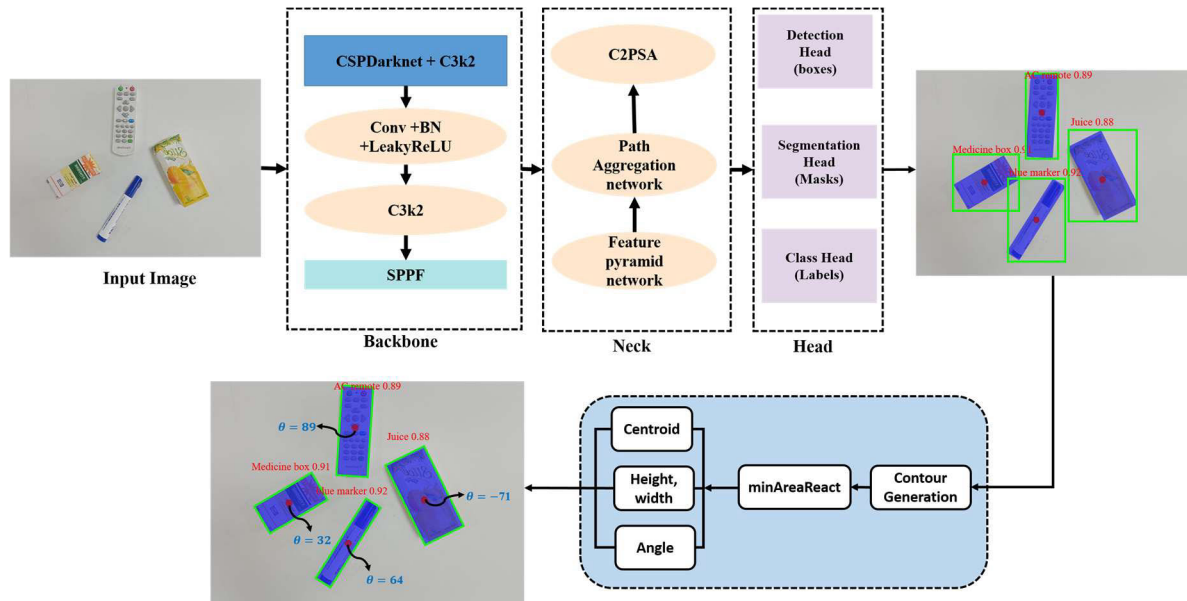


FIGURE 3. Improved YOLOv11-seg model for orientation information. YOLOv11-seg model is post processed by adding additional steps to find out the orientation-related information of objects in the environment.

but serve to evaluate the robustness and scalability of the system. By incorporating a broader range of object types, the system can be tested under realistic conditions, where multiple objects of different shapes, sizes, and textures are present in the same environment.

2) IMPROVED YOLOV11S-SEG MODEL FOR GRASP DETECTION

The proposed system utilizes the YOLOv11s model for real-time object detection within the workspace, employing an Intel RealSense RGBD depth camera positioned on the tabletop in a fixed orientation relative to the Universal robotic arm (UR5e). YOLOv11s model employs CNN layers to perform detection in a single shot. means it produces the object detection and bounding boxes simultaneously in single pass. Renowned for its real-time performance, YOLOv11 excels in applications requiring instantaneous responsiveness, accurately determining the locations of multiple objects within an image. The model's adaptability, available in various sizes, allows customization to our workspace and object recognition goals. Specifically, YOLOv11-seg model, which is the further extension of detection model used for segmentation, it improves the accuracy due to its pixel wise classification capability, which not only predicts objects but gives pixel wise classification. This information is very important for the grasp pose estimation. YOLOv11 is the newest version in the YOLO series by Ultralytics, built on top of YOLOv8. It introduces several architectural changes that improve its performance across many tasks. One of the major changes is the C3k2 block, which is used in place of C2f block, which enhances model efficiency. For better features extraction the model still uses the SPPF (Spatial Pyramid Pooling-Fast) layer but incorporated a new module C2PSA [60], [61]. Due

to these modifications YOLOv11 model can now detect and segment small objects in images with higher accuracy, even in partial occlusion. For the literature it is concluded that YOLOv11s-seg model outperformed other successor models in both accuracy and inference time, which gives us the option to be utilized in assistive grasping system because of higher accuracy, lower inference time, detecting small and occluded objects. As a result, the YOLOv11s-seg model was selected for real-time experimentation in this study. The YOLOv11s-seg model was trained on a Windows 10 system using a GeForce RTX 3060Ti graphics card from NVIDIA. A learning rate of 0.001; a batch size of 16; momentum at 0.937; weight decay set to 0.0005; 500 epochs for the number of iterations; and scaled photos to 640×640 pixels were among the hyper-parameters selected for the training procedure.

Detecting how a robot should grasp an object is crucial for its manipulation abilities. It helps the robot understand where the center of the object is and in which direction it's positioned. However, the YOLOv11s-seg model, which is good at finding object centers, falls short when it comes to figuring out the object's orientation. This becomes a problem for the robot's control module because grabbing an object successfully depends on the robot's end effector or gripper being aligned with the object's orientation. To address this challenge, improved version is developed, which includes the post-processing technique to estimate the object orientation as well as the centroid as shown in Figure 3.

The post-processing steps involve first extracting the mask predicted by the YOLOv11s-seg model for each object in the workspace. These masks are then turned into contours which further create an oriented bounding box on each contour of the object. This oriented bounding box helps to get the orientation of overall object, which is an important parameter

for our grasp pose estimation, which tells us in detail how the object is oriented and positioned. Widely used in computer vision, this technique helps in both localizing the object and estimating its orientation. The final step involves combining the obtained rotated rectangle data with the centroid and orientation information, resulting in a comprehensive prediction. The integration of post-processing techniques fills the gap left by the YOLOv11s-seg model, providing a complete solution that includes both the center points and orientations of detected objects. Hence, the post-processing techniques act as a refinement layer, enhancing the YOLOv11s-seg model's capabilities by introducing a mechanism to understand object orientation.

3) HAND DETECTION AND TRACKING

Handing over the object to the visually impaired person is very challenging task due to safety concerns. An adaptive and robust solution is required to efficiently and safely handover the grasped object to user by detecting and tracking the hand pose [62]. The proposed system uses the bounding box of the detected hand to estimate its centroid and spatial position relative to the robot. Once the YOLOv11s model detects the hand in the RGB frame, the segmentation masks are obtained in frame t denoted by $H_t(x, y)$, and the centroid of the hand region is computed by equation (1);

$$(x_h, y_h) = \left(\frac{1}{N_H} \sum_{H_t} x, \frac{1}{N_H} \sum_{H_t} y \right) \quad (1)$$

where N_H denotes the total number of pixels in the segmented hand region. The depth value corresponding to this centroid is obtained from the RealSense depth camera, providing the 3D position of the hand as $H_p = [x_h, y_h, z_h]^T$. To ensure a safe and stable handover, the algorithm verifies that the user's hand remains stationary before initiating the placement motion. Once stability is confirmed, the target pose for the end-effector is defined based on the current hand position by equation (2);

$$T_{handover} = \begin{bmatrix} R & H_p + z_{off} \\ 0 & 1 \end{bmatrix} \quad (2)$$

where R represents the desired end-effector orientation matrix, and z_{off} is a small vertical offset vector that ensures a safe approach distance before the object is gently handed over to the user. After reaching the pose $T_{handover}$, the robot gradually reduces the offset height in small increments until the gripper aligns precisely with the user's hand position.

Combining the information, hand depth information from the Intel RealSense camera, hand detection from YOLOv11s and trajectory planning, makes the handover framework get the precise, adaptable and safe human robot interaction during handing over process. This approach minimizes contact force and prevents accidental collision with the user's hand while ensuring smooth and reliable object handover.

4) OCCLUSION REMOVAL

One of the major challenges in robotic grasping is the occlusion of objects and mostly it occurs in cluttered environment and assistive robotic grasping scenarios, where the target object is mostly partially occluded with other objects in workspace. Directly attempting to grasp a partially covered object can cause damage to the gripper, the robotic arm, or other objects within the workspace. To tackle this challenge, a natural approach is adopted just like humans do: initially, surrounding object causing the occlusion to target object is first removed by shifting it to safe position in workspace, then efficiently and safely grasp the target object which is completely visible to human now [63], [64]. Several approaches have been proposed for occlusion removal and singulation. Traditional methods employ linear singulation trajectories that push or isolate the target object by applying lateral forces to separate it from clutter [65], [66], [67]. Though, this approach needs several iterations to completely get the target object visible to vision system, which directly affects the overall system latency and efficiency. In human-robot interaction (HRI) contexts, time efficiency is crucial for user comfort and safety. Therefore, to expedite occlusion removal, the proposed system employs a pick-and-place strategy, the non-target object occluding the target is grasped and relocated to a safe position before the robot proceeds to grasp the accessible target object.

The YOLOv11s-seg network performs instance segmentation on RGB frames to generate pixel-wise masks for all detected objects in the workspace. Each segmented region corresponds to a unique class label and a binary segmentation mask $M_j(x, y)$. To incorporate geometric depth information, the aligned depth map $D(x, y)$ from the Intel RealSense D457 camera is utilized. For each segmented object the mean depth value is computed by equation (3):

$$h_j = \frac{1}{N_j} \sum_{(x,y) \in M_j} D(x, y) \quad (3)$$

where N_j denoted the total number of pixels belonging to object j .

Let the target object be denoted with subscript t with centroid as $P_t = (X_t, Y_t)$ with mean depth h_t , and all other objects other than the target object denoted by subscript k . The centroids and the height are denoted by $P_k = (X_k, Y_k)$ with mean depth h_k . The relative depth difference and planar Euclidean distance from center of target object and surrounding objects are computed by equation (4) and (5) respectively.

$$\Delta h_{t,k} = h_k - h_t \quad (4)$$

$$d_{t,k} = \left\| \begin{bmatrix} X_t & X_k \\ Y_t & Y_k \end{bmatrix} \right\| \quad (5)$$

To iteratively check the mask overlap of target object with surrounding objects are computationally expensive and inefficient for real-time operations. To reduce this iterative mask overlap condition, only the closest neighboring objects to the target object are considered for occlusion analysis, while the rest of the objects which are farther away are ignored

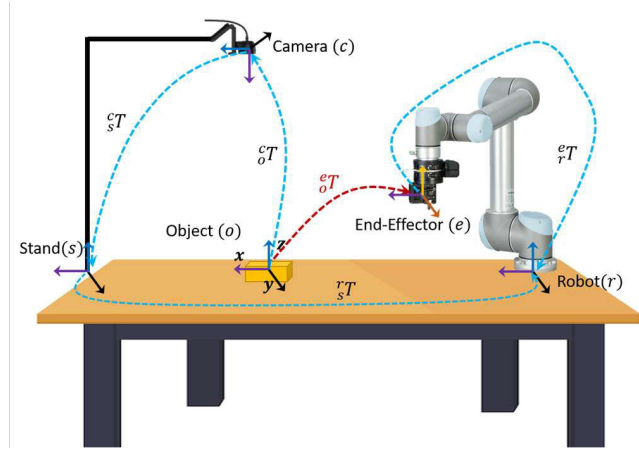


FIGURE 4. Frames transformation from camera coordinates to robot base coordinate systems. This illustrates the object's centroid and orientation-related information transformation from the camera frame to the stand frame and from the stand frame to the robot base frame.

using equation (5). A Euclidean distance threshold is applied, where only the objects at a distance less than or equal to the length of the target object are considered. This approach significantly improves computational cost without affecting the occlusion detection and removal algorithm.

Since the YOLOv11s-seg model provides masks of only the visible parts of objects, the masks of nearby objects may not directly overlap in the segmentation output. Therefore, the segmentation masks are first dilated using morphological operations with kernel size 20 to expand their boundaries slightly. After this boundary expansion, the overlapping condition is checked to detect potential occlusion between objects. An object is considered occluding the target if there is sufficient overlap between the target mask and the neighboring object mask, and if the difference in their mean depths exceeds a predefined threshold. These conditions are expressed by equation (6), and (7), respectively.

$$\frac{|M_t \cap M_k|}{|M_t|} > \tau_o \quad (6)$$

$$\Delta h_{t,k} > \tau_h \quad (7)$$

where M_t , and M_k represent the binary masks of the target and closest neighboring object, respectively, while τ_o and τ_h denote the overlap and height thresholds and in the proposed scenarios $\tau_o \geq 500$ pixels, and $\tau_h \geq 3$ mm.

When both equations (6) and (7) are satisfied, object k is identified as the occluding object. The class ID and class name of this occluding object is retrieved. The coordinates of the occlusion object's centroid are then transmitted to the robot control module, which executes the occlusion removal algorithm. This framework plans a safe trajectory planning to remove occlusion by picking the object causing occlusion and placing it to safe location. Once the target object is completely visible then the robot again plans the new trajectory planning for grasping the target object and then the system hands over the object to the user.

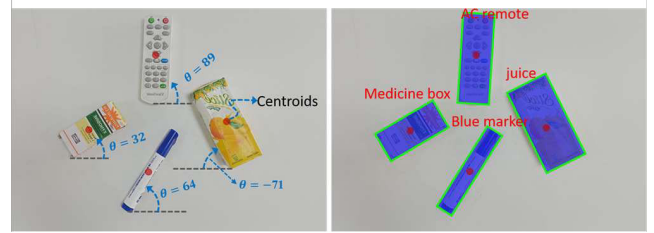


FIGURE 5. Object grasp angle/ orientation estimation. Figure illustrates the orientation-related information of different objects in the camera coordinate system. The angles are being measured about z-axis.

C. VOICE MODULE

The voice module is responsible for the audio input recognition, and conversion of audio to text for better understanding of the algorithm. The voice module is the necessary portion of assistive grasping as it creates a bridge between the user voice input and the robot action. The Voice module is divided into voice recognition, background noise removal, and text classification using machine learning.

1) VOICE RECOGNITION

The core functionality of the proposed system revolves around the voice recognition system, enabling users to easily instruct a robotic arm to locate a desired object within the workspace.

The main purpose of the voice recognition algorithm is to listen and understand voice command, then convert the user voice command to the textual form, which ultimately is sent to the robotic system for further action. The effectiveness of our system hinges on the accuracy and uninterrupted operation of the voice recognition system. However, different people have different language accents and pronunciations which degrade the voice recognition system. To overcome this challenge, different open-source speech recognition models were tested, including the Microsoft Speech API, Sphinxbase support library, CMUSphinx API, and Google Voice Recognition API. The study conducted in [68] highlights that, among these engines, the Google Voice Recognition API stands out as the most reliable with least Word Error Rate (WER) of 9%, as compared to other listed models. On other hand, Sphinx achieved a WER of 37%, and the Microsoft API achieved a WER of 18%. Additionally, the Google Voice Recognition API relies on an internet connection for its operation.

The proposed system involves two-way communication using voice: one is that user can issue a voice command to the robotic system and another communication is when the system sends the voice message containing information regarding the objects in the workspace and when system asks for the voice input when the system is initiated or after completing one task.

Importantly, the system notifies the user in case of an incorrect voice command or the absence of the specified object in the workspace. Uninterrupted voice recognition

system is highly important for the overall system accuracy and reliability to the user. The goal is to develop a user centric system, which takes the voice command from user for quick robotic action and also send the voice message to the user informing about the workspace and status of the robotic arm sequence to update the progress to the user.

2) BACKGROUND NOISE REMOVAL USING BAND PASS WITH SPECTRAL FILTER

A major challenge in voice recognition system is the background noise, which often combines with the voice command of the user, causes the recognition system to degrade their recognition accuracy. These noises can be low, medium and high range but normally in the room or laboratory these noises are the noise of the two-person talking, fan or normal equipment running noise. To enhance the efficiency of the Google Voice recognition system, a pre-processing step is implemented before submitting raw voice data. This pre-processing contains different filters to omit background noise from the voice command of the user. Among different filters, the most promising was the hybrid band-pass with spectral filter to effectively separate the voice command from the background noise. Its effectiveness is evaluated using the Global Signal-to-Noise Ratio (SNR), a metric that gauges the performance of the filter. The results of this evaluation indicate satisfactory outcomes, affirming the band-pass with spectral filter's capability in enhancing the voice recognition system's proficiency [68]. This pre-processing step highlights the significance of tackling the background noise to enhance the system capability to efficiently recognize the user voice command.

3) TEXT CLASSIFICATION USING MACHINE LEARNING

Once the voice command is pre-processed to remove background noise is then passed through Google Speech Recognition API for text conversion. This voice command now in text command undergoes the machine learning based classification model. This classification distinguishes between valid and invalid voice commands. The valid command is associated with the grasping/picking command, while the invalid command is associated with other than grasping command. When the ML algorithm identifies a voice command as valid, it is transmitted to the robotic system. Conversely, if the ML algorithm categorizes the voice command as invalid, the system generates a voice message conveying the message "Invalid command." But before classifying the textual command the ML model is first trained on the custom dataset which contains the text command and labels as valid "1" and invalid "0". Five different machine learning models were considered: logistic regression, Support Vector Machine (SVM), decision trees, and random forest. Among these five ML models, SVM emerged as the optimal choice, and recorded the highest accuracy at approximately 97.93% compared to the other models. This real-time ML based text classification increases the intelligent safety system,

which only allows the valid/grasping related commands to the robotic controller for action.

D. ROBOT CONTROL MODULE

To grasp the targeted object, the positional and orientation-related information of the object is passed on to the robot using transformation matrices. This matrix plays a pivotal role by converting information captured by the camera to align with the robot base (global coordinate system), ensuring accurate object manipulation. The overall scenarios related to the transformation matrices are shown in Figure 4.

The transformation matrix carries information related to the robot end-effector with respect to the global coordinate system (robot base). r_oT transformation matrix which transforms the end-effector information into a robot base coordinate system is known since all the joint angles can be calculated using inverse kinematics. Next, the transformation matrix e_oT which represents the object position and orientation with respect to the end-effector coordinate is unknown. Moreover, from Figure 4 one can write the homogeneous transformation matrix r_oT which transforms the object information into the robot base coordinate system as follows.

$${}^r_oT = {}^r_eT {}^e_oT \quad (8)$$

where the transformation matrix r_eT can be written as follows

$${}^r_eT = \begin{bmatrix} {}^rR_o & {}^rP_o \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} C_\theta & -S_\theta & 0 & {}^rP_{x_o} \\ S_\theta & C_\theta & 0 & {}^rP_{y_o} \\ 0 & 0 & 1 & {}^rP_{z_o} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

where rR_o and rP_o denote the orientation and position-related information of the object in robot base coordinate system, respectively. As all the objects are placed on a tabletop, the objects have only rotation θ about z-axis.

Figure 5 depicts the estimation of each object's orientation about z-axis. An angle less than 90 degrees is being considered as positive, and an angle exceeding 90 degrees is computed by subtracting it from 180 degrees, indicating its supplementary angle with a negative value. These supplementary and complementary angles are then used in the rR_o which transforms the orientation information of each object to the robot base frame. In equation 8, e_oT transformation matrix is unknown. To calculate the e_oT , we can write homogeneous transformation matrices equation from Figure 4 as follows.

$${}^r_eT {}^e_oT = {}^r_sT {}^s_cT {}^c_oT \quad (10)$$

where the transformation matrices r_sT and s_cT are constant. s_cT transforms the camera coordinate system into stand coordinate system and c_oT transforms the stand coordinate system into robot base coordinate system as shown in Figure 4. Moreover, the c_oT transformation matrix transforms the information of objects (position and orientation) into camera coordinate system. The equation 10 can be written as follows:

$${}^e_oT = ({}^r_eT)^{-1} {}^r_sT {}^s_cT {}^c_oT \quad (11)$$

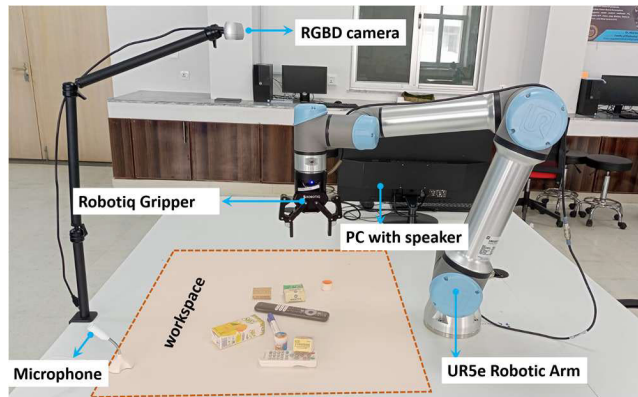


FIGURE 6. Experimental setup. The Figure illustrates the overall setup used for the validation of the proposed intelligent assistive grasping system which includes the RGBD camera for the vision module, microphone for the voice module, and the UR5e robotic arm for the robot control module.

Finally, one can find the oT by substituting equation 8 back into equation (8). The transformed information is then used by the inverse kinematics of the UR5e to determine each joint position of the robot in joint space and to ensure a non-singular configuration during object grasping and handover to the user. An additional safety check is implemented to prevent the robotic arm from moving to extreme positions or too close to the depth camera and its frame, thereby avoiding any potential collision between the robot and the camera setup hardware. All computations related to the transformation of object location and orientation are executed within the Python API. One important task which also involves the robot control module is the object handover to the user hand according to the hand position. This action needs a careful trajectory planning to ensure the safety of the user as well as robot. All of these transformations and motions are performed in Python programming language with the UR5e robotic arm's API to have smooth and precise motion in both grasping and handing over tasks. The Universal Robot (UR5e) is broadly known robotic manipulator known for its precision and adaptability in both research and industrial applications. Detailed specifications are provided in Appendix.

IV. RESULTS

A. EXPERIMENTAL SETUP

To validate the performance of the proposed assistive grasping system, each module is tested and evaluated both separately and overall, as a single assistive grasping system. Figure 6 shows the overall experimental setup for validating the intelligent robotic grasping manipulator for visually impaired person. The setup consists of an Intel RealSense RGB-D depth camera, a microphone, and a Universal Robots UR5e manipulator, all integrated through a unified control interface. The RGB-D camera continuously captures synchronized RGB and depth data from the workspace, providing accurate object localization and 3D spatial information. The workspace is the orange dotted rectangular area

where UR5e robotic arm can safely reach and grasp the target object, and this area must be in the camera view to capture all the object placed in it. To ensure user safety and prevent mechanical damage, the robot is programmed to avoid executing actions when the detected object lies beyond the defined workspace limits, thereby protecting the arm's joint actuators and motors.

All modules were integrated on a desktop workstation running Ubuntu 22.04.5 LTS (64-bit) with Linux kernel 6.8.0-85-generic. The computational platform is powered by a 12th Generation Intel Core i9-12900K processor (24 cores, 16 threads) and 32 GB DDR4 RAM, equipped with Intel AlderLake-S GT1 (UHD Graphics 730) GPU. This system configuration provides sufficient computational capacity for real-time visual perception, speech processing, and robotic control tasks, ensuring smooth and low-latency operation during experiments.

For comparison among YOLOv8s, YOLOv9c, and YOLOv11s, all the models were trained on a dedicated workstation running Windows 10 (64-bit) as operating system. The training environment was powered by a 13th Generation Intel Core i7-13700 processor, 32 GB of RAM, and dual GPUs comprising Intel UHD Graphics 770 (integrated) and an NVIDIA GeForce RTX 3060 (12 GB VRAM) discrete GPU.

Safety Measures: To ensure user and system safety, both hardware and software safety checks were implemented. The UR5e Cobot features an integrated force/torque sensor that enables real-time collision detection and compliant motion control, it stops movement upon detecting excessive external force. In parallel, workspace constraints were embedded within the control algorithm; if a commanded target lies outside the safe workspace, the robot automatically stops execution and requests a new command. During handover, the system maintains a safety buffer distance from the user's hand and gradually reduces approach height using depth feedback from the RealSense camera. These layered safety mechanisms prevent overreach, collisions, and unintended contact, ensuring safe and reliable human-robot interaction.

B. SYSTEM MODIFICATION FROM USER PERSPECTIVE

Different sighted users were asked to test and evaluate the proposed assistive grasping system. The first recommendation was to incorporate the more diverse picking and grasping phrases to perform grasping action, therefore significant amount of dataset was collected for different input phrases. The second recommendation was to incorporate a filtering system, whether the impractical command or not related to grasping is provided the system should identify that command and not perform action on it. Third suggestion was to activate and de-activate the robotic system also with the voice command, if the user doesn't want to perform action for some time the robot should be in the sleep mode. Users also wanted to improve the handing over system, initially a pre-defined position was used for handing over the object that lacked the adaptability and robustness of the system. One drawback

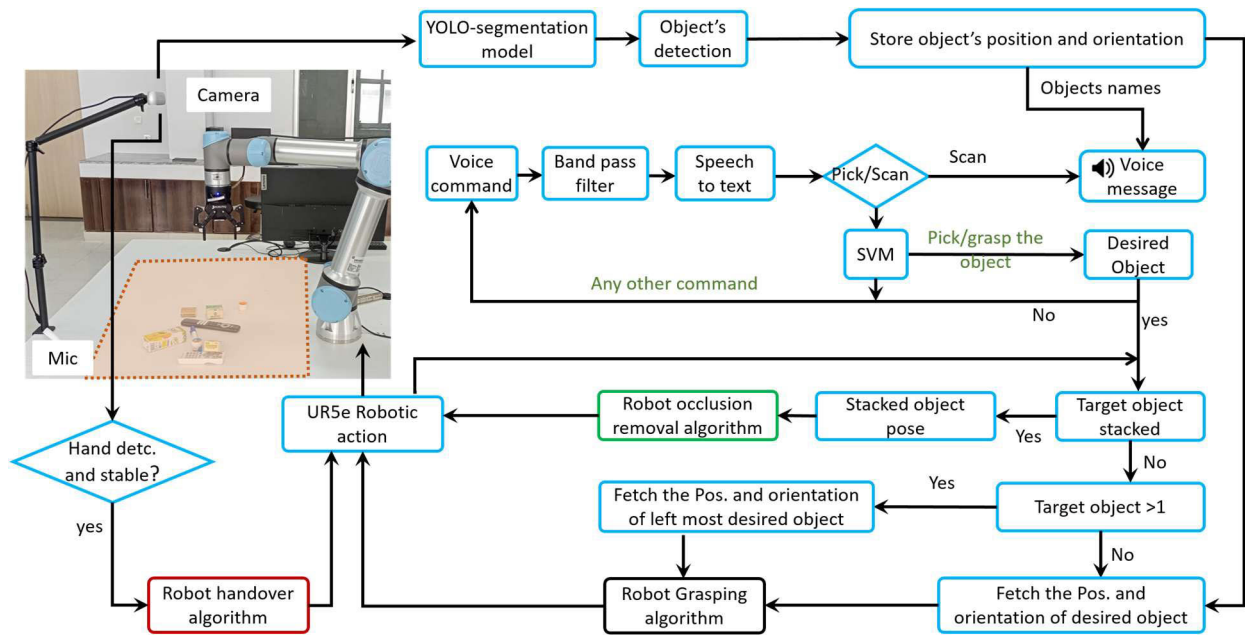


FIGURE 7. Flow diagram of the overall system. Figure illustrates how all three modules are integrated with each other. This shows that the system has two inputs, the vision module and the voice module, and the output would be the robot control module.

during the initial testing was the mixture of the background noise with the voice command of user, which the speech to text algorithm finds difficult to perform due to the background noise. According to the suggestions proposed by the initial testing, the improved version included the following improvements.

1. Different voice phrases for the voice command to the robot with the background noise removal filtration.
2. Another safety condition is to predict whether the voice input is related to the grasping or picking up the object or not using the SVM (support vector machine) algorithm.
3. Activation and de-activation of the robotic system using voice command for idle condition.
4. Incorporation of the band pass with spectral filter to efficiently remove background noise from the voice command.
5. Hand detection, tracking and stability check for safely handing over the object to user according to position and height.

C. EXPERIMENTAL EVALUATION OF EACH MODULE

The flow diagram of the experimental setup with the voice module, machine vision module, and robot action module is shown in Figure 7. Once the system is initialized, the camera scans the workspace, captures the image of the workspace, and passes this captured image from the trained YOLOv11s-seg model. The detected objects with their centroids and orientations are stored in memory for further usage, and a voice message is sent to the user: “How may I help you?”. The user can then choose between two options: (1) the system scans the environment and speaks the names of the detected objects, or (2) the system picks/grasps the desired object requested by the user. The system will then wait for the

voice command from the visually impaired person. Before passing this voice command to Google API for speech-to-text conversion, background noise may interfere with accurate recognition and lead to incorrect transcription. To address this, a band-pass filter is used to remove background noise so that the speech-to-text conversion is performed accurately. If the user’s voice command is “scan the environment/table”, the system will send a voice message listing all the detected objects. For commands other than scanning, the input is passed through a machine learning algorithm to classify the text as either a valid or invalid command. A valid command means the visually impaired person wants to pick an object, while an invalid command means the user has given a voice command that does not correspond to object-picking. Once a valid command is received, the object the user wants to pick is specified, and the centroid and orientation information is retrieved from the stored memory for application. If the target object is not found in the scanned environment, the system continues scanning until the target object is detected.

There are three conditions when the target object is detected:

1. If the target object is unique/single, the object’s centroid and orientation information are passed to the UR5e robot for the grasping algorithm.
2. If there are multiple instances of the target object, the leftmost target object’s poses are sent to the grasping algorithm.
3. If the target object is stacked (i.e., another object is placed above it), the system will remove the object causing the occlusion using the robot occlusion removal algorithm.

Once the target object is grasped, the machine vision module will wait until the hand is detected. The hand should remain stable to allow the object to be handed over safely

in the environment. After this, one cycle of the system is completed, and it is ready for another voice command to start a new loop.

Algorithm 1 Assistive Grasping for Visually Impaired

```

1: %Initialize system modules:
   YOLOv11s-seg model
   Intel RealSense D457 camera
   UR5e robotic arm
2: while system is active do
3:   Capture RGB frame and aligned depth map
4:   Results = YOLOv11s_segmentation(RGB frame)
5:   for each Object. Mask [i] in Results do
6:     Store position[i], orientation[i], and class_name[i]
7:   end for
8:   Voice_Command = CaptureVoiceInput()
9:   Filtered_Signal = Hybrid_BandPass_spectral_Filter(Voice_Command)
10:  Text_Command = SpeechToText(Filtered_Signal)
11:  if Text_Command == "Scan" then
12:    Display and Aimounce list of detected objects
13:  continue
14:  end if
15:  Conunand_Label = SVM_Classifier(Text_Command)
16:  if Command_Label = 1 then % 1 = grasping/pkk, 0= no grasping
17:    Desired_Object = ExtractObjectName(Text_Command)
18:    if Desired_Object G Detected_Objects then
19:      Target_object ← Desired_Object
20:      Pose Target ← GetPose(Target_Object)
21:      Compute mean depth  $h_t$  and centroid ( $X_t, Y_t$ )
22:      for each object  $k \pm$  Target_object do
23:        Compute  $\Delta h_{tk} = h_k - h_t$ 
24:        Compute  $\Delta_{overlap} = \frac{|M_t \cap M_k|}{|M_t|}$ 
25:        if  $\Delta_{overlap} > \tau_o$  and  $\Delta h_{tk} > \tau_h$  then
26:          Occluder = Object k
27:          Move Occluder to safe_position using UR5e
28:        end if
29:      end for
30:      if CountSimilar_Objects(Target_Object) > 1 then
31:        Pose_Target = GetLeftMostObjectPose(Target_Object)
32:      end if
33:      Plan collision-free trajectory to Pose_Target
34:      Move UR5e to pre-grasp pose
35:      Close gripper to grasp object securely
36:      Lift object to safe height
37:      UR5e_Action("Grasp Complete")
38:      if HandDetectedAndStable() == TRUE then
39:        Move UR5e to handover position
40:        Open gripper to release object
41:        UR5e_Action("Handover Complete")
42:      end if
43:    else
44:      PlayVoiceMessage("Requested object not found.")
45:    end
46:  else
47:    Play VoiceMessage( "Invalid Command for Grasping")
48:  end if
49: end while

```

The complete algorithm for assistive grasping for visually impaired person, which consists of voice recognition/classification, grasp pose estimation and safe handing over to visually impaired is summarized in Algorithm 1. The individual results of the three modules are discussed in detail in the following sections.

1) BACKGROUND NOISE REMOVAL AND VOICE RECOGNITION

Human ears can perceive frequencies ranging from 20 Hz to 20 kHz; however, understanding speech becomes difficult in noisy environments, a challenge shared by voice-recognition systems that rely on microphone inputs. To enhance recognition accuracy and suppress background interference, a band-pass spectral filtering method was implemented. First,

TABLE 3. Performance of voice recognition system under varying background noise levels, showing the effect of noise on recognition accuracy and latency.

Noise level	voice commands	Correctly recognized	wrong recognized	Voice Recognition Accuracy (%)
Low noise (Quiet room, <35 db)	30	30	0	100%
Medium noise (office/Fan Noise, ~50db)	30	28	2	93.3%
High noise (Background conversation/ Tools~70db)	30	26	4	86.7%
Overall average	90	84	6	93.5%

a digital band-pass filter with cutoff frequencies between 80 Hz and 8 kHz was applied to preserve the useful speech components while attenuating low-frequency hums and high-frequency noise. Subsequently, spectral noise reduction using an adaptive gating-based algorithm was performed. The traditional filters use constant thresholds to filter out the noise while the spectral filters estimate the noise profile from non-speech region in voice and then choose cutoff frequency to filter out the noise below dynamic noise floor, which gives the clean voice command to the recognition system.

The detailed testing was performed using 30 different voice commands under three different levels of noise conditions (low, medium, and high) to further validate the system effectiveness. The results are summarized in Table 3, the system achieved 100% accuracy in low-noise environments (<35 dB), correctly recognizing all 30 commands. In medium-noise conditions (~ 50 dB), 28 out of 30 commands were accurately converted to text, yielding 93.3% accuracy, while in high-noise conditions (~ 70 dB), 26 out of 30 commands were recognized with 86.7% accuracy. The overall recognition accuracy across all tests was 93.5%, demonstrating that the combined band-pass and spectral-gating approach effectively suppresses both broadband and speech-overlapping components.

To visualize the improvement, Figure 8 presents a comparison between the original noisy voice signal (red) and the filtered signal (blue). The results clearly illustrate significant noise attenuation and preservation of speech intelligibility. Six representative commands “Pick up the red marker,” “Pick up the blue marker,” “Pick up the medicine box,” “Pick up the black marker,” “Pick up the solder paste,” and “Pick up the TV remote” were selected for detailed evaluation under constant background noise. The processed waveforms confirm that the proposed filtering pipeline substantially enhances clarity and consistency, enabling more reliable voice command recognition even in acoustic challenging environments.

2) MACHINE LEARNING BASED TEXT CLASSIFICATION, PERFORMANCE EVALUATION AND COMPARISON

To improve the accuracy of the text classification system for assistive robotics system, the dataset was expanded from

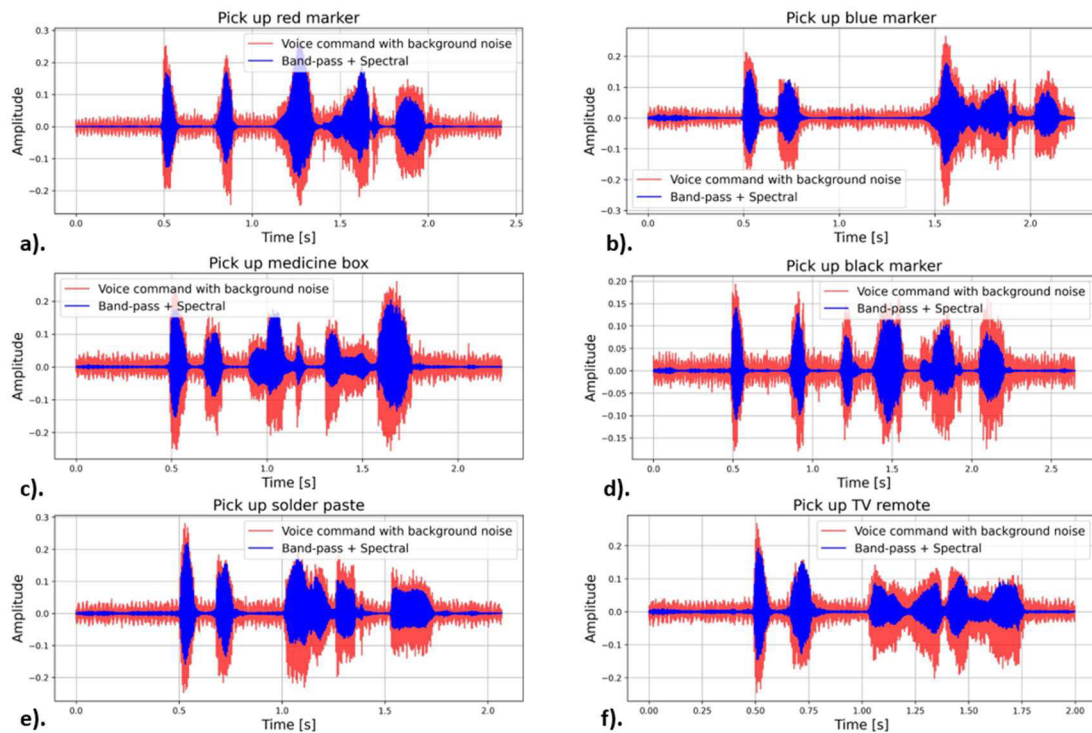


FIGURE 8. Background noise removal results for six voice commands. Each subplot shows the original noisy audio signal (blue) and the filtered signal after noise removal (red). The voice commands are indicated in the subplot titles: (a) Red marker, (b) Blue marker, (c) Medicine box, (d) Black marker, (e) Solder paste, and (f) TV remote.

1051 sample using advanced paraphrasing and augmentation techniques to achieve 3050 voice commands with labels. This included (1) natural language diversity with filler words and human-like hesitations (e.g., “uh pick the remote”), (2) contextual variations such as spatial references (“grab the key near the table”), (3) typo simulation to mimic speech-to-text artifacts, and (4) harder negative samples involving polite negations (e.g., “please don’t move the tap”), which closely resemble positive commands in structure. This augmentation significantly improved dataset generalization and model robustness against linguistic ambiguity and noise. Each voice command which is the text command in dataset is either labelled as 1 (valid/grasping command) or 0 (irrelevant command). For example, positive commands are “pick up the bandage,” “bring me the TV remote,” and “robot grasp the solder paste,” and on the other hand negative samples included “don’t pick the juice” and “leave the black marker.” The dataset was balanced with 1517 positive and 1533 negative examples ($\approx 50:50$ split), as shown in Table 4.

Each data sample in the dataset was pre-processed with lowercasing, punctuation and full stop removal. It is then converted into features into numerical representation using Term Frequency–Inverse Document Frequency (TF-IDF) method, which extracts the importance of each keyword and matches with labels. Five different ML models were trained, named as: Support Vector Machine (SVM), Logistic Regression (LR), Random Forest (RF), Decision Tree (DT), and K-Nearest Neighbors (KNN), and evaluated using 5-fold

TABLE 4. Voice command classification dataset structure.

Example command	Label	Meaning
Grab the remote	1	Valid grasp command
Handover the key	1	Valid grasp command
Don’t pick the juice	0	Irrelevant/no action
Please avoid picking the blue marker	0	Irrelevant/no action
Fetch me TV remote	1	Valid grasp command
Robot, that bandage is off the limits	0	Irrelevant/no action

Total samples: 3050, Label 1: 1517(49.73%), Label 0: 1533 (50.26%)

stratified cross-validation to confirm reliable performance estimation across data splits. The quantitative performance comparison is summarized in Table 5. The Linear SVM achieved the highest mean accuracy of 97.93%, precision of 0.975, recall of 0.983, F1 score of 0.979, and an AUC of 0.997, confirming its superior discriminative capability for grasp-related command classification. Logistic Regression and Random Forest followed closely, achieving accuracies of 96.65% and 97.15%, respectively, with AUC values above 0.99. The Decision Tree model yielded a noticeably lower accuracy of 87.15% and AUC of 0.871, indicating a weaker ability to generalize to unseen variations. The KNN classifier achieved a balanced performance with 92.23% accuracy and AUC = 0.977, showing moderate generalization capability.

TABLE 5. Text Classification evaluation for different machine learning models (5-Fold cross validation).

Model	Accuracy (%)	Precision	Recall	F1 score	AUC
SVM	97.93	0.975	0.983	0.979	0.997
Logistic regression	96.65	0.956	0.977	0.966	0.993
Random forest	97.14	0.969	0.973	0.971	0.996
Decision Tree	87.14	0.867	0.876	0.871	0.871
KNN	92.229	0.912	0.933	0.922	0.977

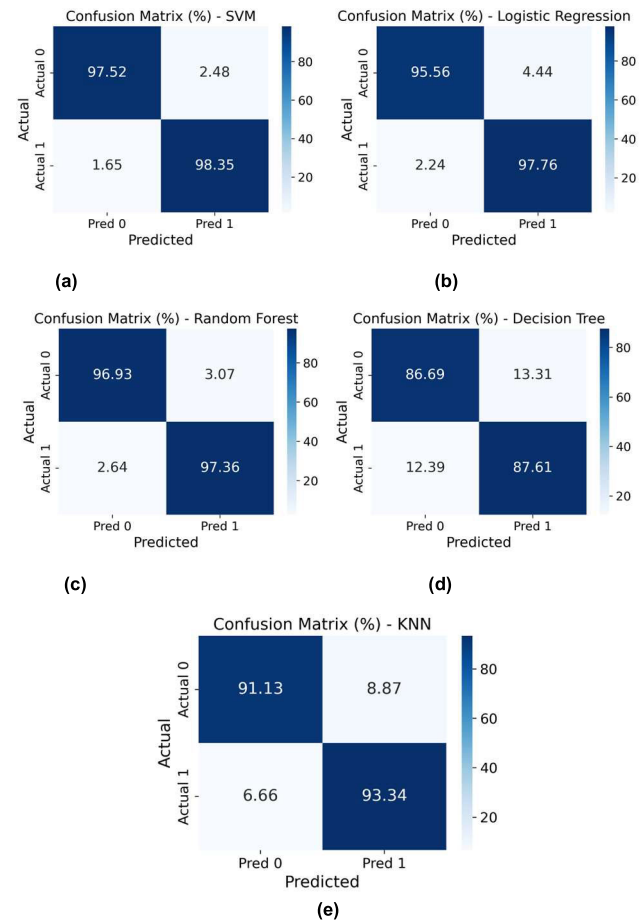


FIGURE 9. Confusion matrices showing classification performance of five models, (a). SVM, (b). Logistic regression, (c). Random Forest, (d). Decision Tree, and (e). KNN.

The confusion matrices further illustrate the classifiers' ability to correctly identify each command class shown in Figure 9. SVM achieved the highest class-wise accuracy, correctly classifying 97.52% of negative commands and 98.35% of positive commands, indicating excellent discriminative power. Random Forest also performed strongly, with 96.93% and 97.36% correct predictions for the respective classes. KNN showed balanced performance with 91.13% and 93.34% accuracy for negative and positive commands, while Decision Tree struggled comparatively, particularly in distinguishing negative commands, with only

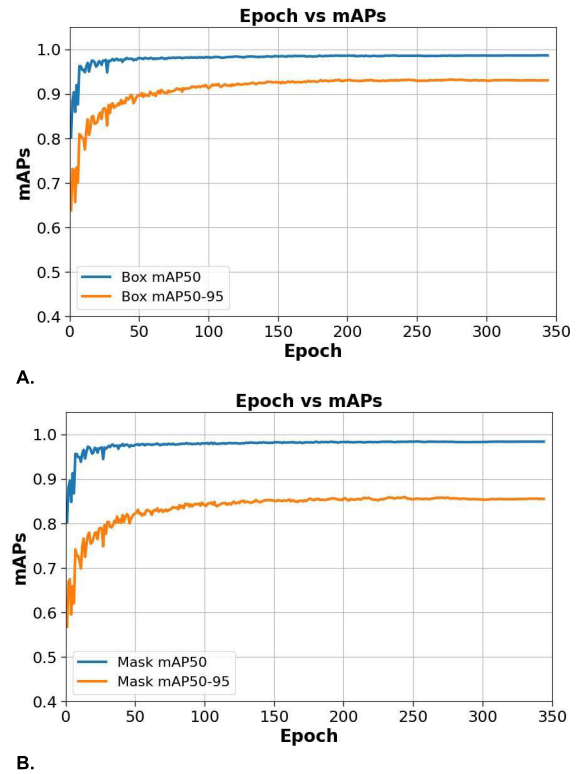


FIGURE 10. Performance evaluation (a) Epoch vs mAP@50, and mAP@50-95 graph for Bounding box (b) Epoch vs mAP@50, and mAP@50-95 graph for Mask.

86.69% correct predictions. Overall, the confusion matrices confirm that SVM provides the most reliable and robust classification across both classes, supporting its selection as the primary model for the voice command recognition module.

3) PERFORMANCE EVALUATION OF OBJECT DETECTION AND SEGMENTATION MODEL

Figure 10 illustrates the model's training performance, showcasing the evaluation of mAPs over different epochs for bounding boxes and the masks, respectively. Figure 10a depicts the mAPs for bounding box predictions at an IoU threshold of 0.50 (mAP50) and at IoU threshold of 0.5-0.95 (mAP50-95). The mAP50 graph rapidly increased from an initial low value to 0.986, reflecting excellent precision and recall. While the mAP 50-95 for box also rises quickly and stabilizes, though at a slightly lower level due to the more stringent criteria at 0.933. Similarly, Figure 10b, presents the mAP for segmentation masks at an IoU threshold of 0.50, mirroring the trend seen in the bounding box mAP50 and reaching 0.983. While the mAP for segmentation masks over varying IoU thresholds (mAP50-95), showing a consistent increase followed by stabilization at 0.855.

These results suggest that the model has effectively learned to detect and segment objects with high accuracy, maintaining strong performance across different evaluation thresholds.

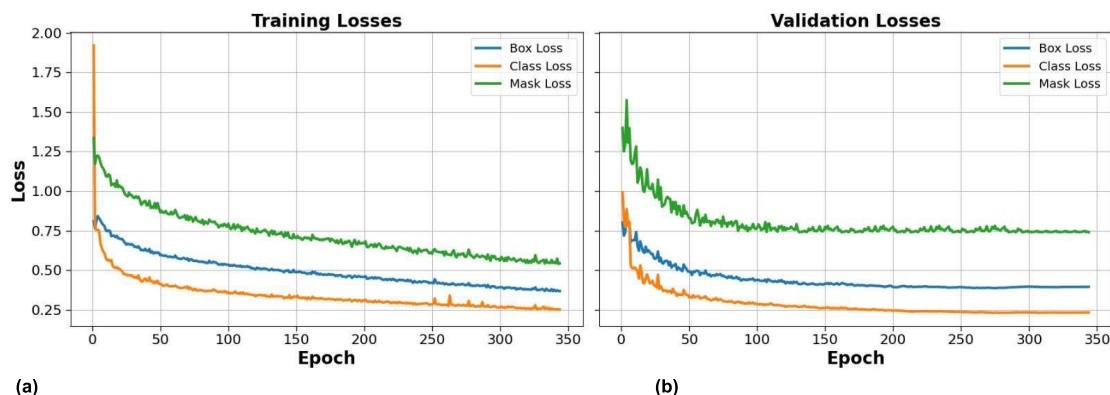


FIGURE 11. Epoch vs Loss graph for YOLOv11s-seg model. During model training, the loss at each epoch is plotted for the box, class and mask separately.

Figure 11 illustrates the training loss curves of the YOLOv11s segmentation model for 500 epochs. However, the Training was scheduled for 500 epochs but early stopping terminated training at 344 epochs. The graphs are further categorized into three losses: Box loss (blue), Class Loss (orange), and Mask loss (green). Box loss measures the accuracy of the predicted bounding boxes around the objects; class loss evaluates the model's ability to correctly classify the detected objects, and mask loss assesses the accuracy of the correctly predicted segmentation masks. Initially, all losses were high, indicating the model started learning. After some initial epochs each loss started to decrease, showing that the model is improving. The steep drop in class loss early on suggests the model quickly learns to classify the objects, while the box and mask losses decrease more gradually, indicating a more complex training process for precise bounding box predictions and segmentation.

The model was evaluated on a dataset containing thirteen object types, achieving notable performance with a mean Average Precision (mAP) of 0.933 for bounding boxes and 0.882 for segmentation masks at IoU thresholds ranging from 0.50 to 0.95. This demonstrates the model's effectiveness in both identifying and segmenting objects accurately. The model performed well across various orientations of objects, highlighting its robust segmentation capabilities. Table 6 shows the Average Precision (AP) metrics for both bounding boxes and masks for each class. The "Box" column includes AP values at an IoU threshold of 0.5 and averaged over thresholds from 0.5 to 0.95, while the "Mask" column provides similar metrics for segmentation masks. Higher AP values for both bounding boxes and masks across different classes indicate the model's strong detection and segmentation capabilities. Although the AP values decrease with stricter IoU thresholds, the model maintains good performance. The trend of decreasing AP values with increasing IoU thresholds is expected, and our model reflects this pattern. Notably, bounding box AP is generally higher than mask AP due to the greater accuracy required for segmentation masks, resulting in lower AP values for masks compared to

TABLE 6. Average Precision at IoU 50, and 50-95 for bounding boxes and masks across different classes.

Names All classes	Box		Mask	
	AP50	AP50-95	AP50	AP50-95
AC remote	0.995	0.969	0.995	0.942
TV remote	0.994	0.986	0.994	0.951
bandage	0.988	0.942	0.988	0.885
black marker	0.956	0.883	0.948	0.77
blue marker	0.967	0.915	0.951	0.789
juice	0.979	0.966	0.979	0.937
key	0.995	0.945	0.995	0.793
medicine box	0.991	0.949	0.991	0.906
plaster paste	0.995	0.97	0.995	0.933
red marker	0.979	0.924	0.97	0.807
solder paste	0.993	0.88	0.994	0.814
tap	0.987	0.869	0.987	0.773
whitener	0.995	0.925	0.995	0.818
All classes	0.986	0.933	0.983	0.882

bounding boxes at IoU thresholds from 0.50 to 0.95. We focus on reporting the AP metrics for each class rather than presenting the precision-recall curves and confusion matrices. The AP metrics provide a clear and concise summary of the model's performance across each class with range of IoU thresholds (IoU at 50, and 50-95), which is particularly useful for comparing model accuracy in object detection and segmentation tasks. Moreover, the information provided by PR curves and confusion matrices is inherently included in the AP metrics. Therefore, the AP metrics inherently reflect the detailed performance characteristics that would otherwise be illustrated by PR curves. The confusion matrix on the test set is included in appendix.

4) COMPARISON WITH SOTA

To validate the performance and efficiency of the proposed YOLOv11s-seg model, a comparative evaluation was

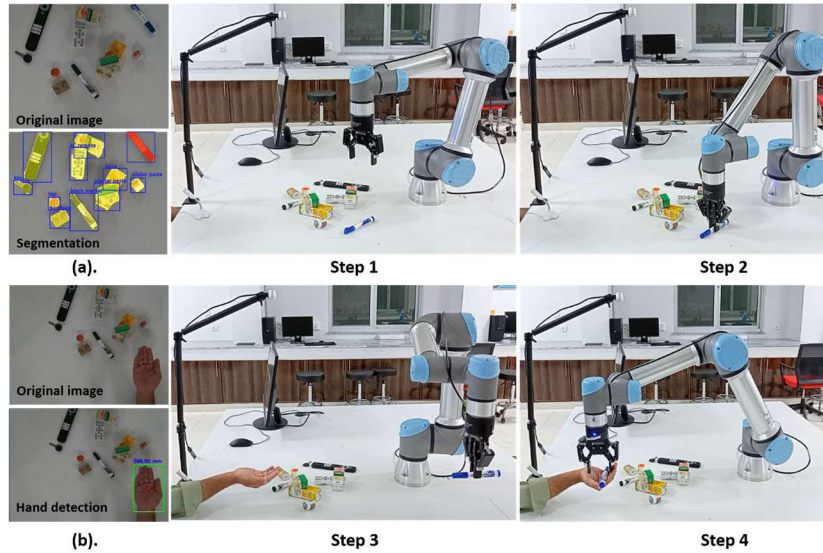


FIGURE 12. Voice command “Pick up the blue marker and the Sequence of robot operation to grasp the objects and handing over to the user while detecting and tracking the hand. a). target object detection and grasping action, and b). robust handover to the user with hand detection and tracking.

TABLE 7. Comparison of YOLOv11s-seg with state-of-the-art YOLOv8s-seg and YOLOv9c-seg model in terms of model complexity, detection and segmentation accuracy, inference time and model size.

Metric	YOLOv8s-seg	YOLOv9c-seg	YOLOv11s-seg
Parameters (M)	11.78	27.63	10.07
GFLOPs	42.5	147.6	32.8
Box Precision (P)	0.983	0.987	0.987
Box Recall (R)	0.978	0.977	0.978
Box mAP50	0.984	0.985	0.986
Box mAP50-95	0.924	0.932	0.933
Mask Precision (P)	0.983	0.987	0.987
Mask Recall (R)	0.975	0.977	0.978
Mask mAP50	0.980	0.983	0.983
Mask mAP50-95	0.848	0.885	0.882
Inference Time (ms/img)	4.3	14.3	4.2
Model Size (MB)	23.9	56.3	20.6

carried out against two recent state-of-the-art segmentation frameworks: YOLOv8s-seg and YOLOv9c-seg, under identical dataset, training, and evaluation settings. The quantitative results, summarized in Table 7, demonstrate that YOLOv11s-seg achieves comparable or superior detection and segmentation accuracy while significantly reducing computational complexity and model size.

In particular, the proposed model attains a box mAP_{50-95} of 0.933 and a mask mAP_{50-95} of 0.882, marginally surpassing YOLOv8s-seg and closely matching YOLOv9c-seg. Despite its compact architecture with only 10.07 M parameters and 32.8 GFLOPs, YOLOv11s-seg maintains identical

precision (0.987) and recall (0.978) for both bounding boxes and masks compared with YOLOv9c-seg. The model also achieves the lowest inference latency of 4.2 ms per image, compared to 4.3 ms for YOLOv8s-seg and 14.3 ms for YOLOv9c-seg, corresponding to a $3.5\times$ reduction in computational cost relative to YOLOv9c-seg. Additionally, its compact file size (20.6 MB) facilitates deployment on embedded or edge devices where computational and memory resources are constrained.

These results collectively demonstrate that YOLOv11s-seg delivers an optimal balance between accuracy, efficiency, and real-time performance, making it particularly well-suited for assistive robotic grasping applications.

D. REAL TIME TESTING OF PROPOSED ASSISTIVE GRASPING SYSTEM

To validate the performance of the proposed assistive grasping for the visually impaired, an experimentation was performed with humans in a loop in our laboratory. Proposed system testing was performed on different human subjects and evaluated for the performance of the assistive grasping system. The experimental setup is already shown in Figure 6.

Different daily-life objects were used to train the detection and segmentation model. The vision module is able to detect and segment all the objects present in the workspace, the voice recognition module is able to recognize and convert speech to text successfully with and without background noise, and the robot control module is able to locate the target object’s location as well as its orientation using transformation matrices as explained in equations (8) to (11). The

experimentation showed promising results with the physical 6DOF robot (Universal Robot UR5e). The sequence of operations included: “pick up the blue marker”, “pick up the TV remote”, and “pick up the AC remote”. Figures 12–14 show the experimentation with robot actions using human voice commands. In Figure 12, the voice command was to pick up the blue marker. The vision module scanned the workspace, and the top image in Figure 12a shows the original image captured from the Intel RealSense depth camera mounted above the workspace using a fixed camera stand. The bottom image shows the results obtained from the YOLOv11s segmentation model, where the target object is highlighted with a red mask and all other objects with a yellow mask. Steps 1–2 show the robot’s actions while picking the target object (blue marker). Once the target object was grasped, the machine vision system waited for stable hand detection. Figure 12b shows the original and hand detection images from the depth camera, and steps 3–4 show the robotic arm handing over the blue marker to the user safely and efficiently on the palm of the hand.

Similarly, in Figure 13, the voice command was “pick up the TV remote”. The original and segmentation images are shown in Figure 13a, where the target and other objects are represented in red and yellow, respectively. Steps 1–2 show the robotic arm picking up the TV remote. Figure 13b shows the original and hand detection images, while steps 3–4 show the handover of the target object to the user’s palm precisely according to the hand location and height.

In another case, the voice command was “pick up the AC remote”, where an interesting situation occurred: the target object (AC remote) was under another object (a medicine box), shown in the red circle in Figure 14a. This figure shows the original and segmentation images. The algorithm first removed the object stacked above the target (medicine box) and then attempted to pick up and hand over the target object to the user. Steps 1–2 in Figure 14 show the robotic action of picking up the stacked object (medicine box) and placing it in the lower-left corner of the camera view. Then, in Figure 14b, the original and segmentation results are again shown, highlighting the target and other object masks. Steps 3–4 show the robotic action of picking up the target object (AC remote). Once the AC remote was grasped, the hand was moved continuously from right to left to evaluate whether the algorithm waited for hand stabilization or handed over the object immediately after capturing the first image frame from the depth camera. The robot waited as long as the hand was moving, and as soon as the hand became stable, the handover algorithm was triggered. Figure 14c shows the original and hand detection images, and steps 5–6 show the handover of the AC remote to the user efficiently. The hand position and height differed in all these situations, and the algorithm successfully handed over the object with precision.

The overall performance of the proposed system for visually impaired people was satisfactory. The methodologies and successful integration of different modules with real-time

implementation can easily be extended for the development of assistive technologies for people with other disabilities.

V. OVERALL SYSTEM LATENCY AND PERFORMANCE EVALUATION

A. COMPLETE ASSISTIVE GRASPING SYSTEM LATENCY ANALYSIS

Table 8, details the end-to-end latency of the complete assistive manipulation process, measured from the instant a voice command is spoken to the final handover of the object to the user. The total latency includes speech duration, speech-to-text recognition delay, command classification, and robot execution time. Across 20 representative test commands, the system achieved an average speech duration of 3.22 ± 0.31 sec, recognition latency of 2.27 ± 0.56 sec, and total command processing time of 5.48 ± 0.63 sec. The robot required an average of 31.35 ± 1.01 sec for grasping and 5.84 ± 0.54 sec for handover, resulting in an average total robot action time of 37.19 ± 1.43 sec and an overall interaction time of approximately 42.67 ± 1.47 sec from voice input to task completion. These results demonstrate that the system achieves responsiveness with consistent timing behavior across multiple trials. Minor variations in total execution time primarily arise from different lengths of voice commands, target object and hand position in the workspace or target object occlusion.

B. GRASP SUCCESS RATE EVALUATION

Figure 15 presents the grasp success and failure rates for 13 common objects tested in the assistive manipulation experiments, each evaluated over ten independent trials. The results were recorded after the voice command was successfully recognized, converted to text, and classified as a valid grasping command. This evaluation highlights the reliable grasping performance of the system, which ultimately is connected to vision module in generating accurate masks for the target object, estimation of grasp poses, and accurate implementation of transformation of information from the camera frame to the robot base frame. Overall, a total of 130 grasping attempts were performed on each object with 10 tests. The proposed framework successfully completed the grasp for 123 attempts with mean success rate of approximately $94.7\% \pm 7.76\%$, while the unsuccessful grasp was 7 with 5.3% failure rate. Higher failures were observed for the key and tap due to lower height and small size respectively. Some failures also occurred for the bandage, solder paste and whitener, and the reasons for their failures were object deformability, small size, and sometimes in accurate segmentation, respectively. On the other hand, all rigid and medium-sized objects, including remotes, markers, the juice box, and medicine box, achieved a 100% success rate. These results demonstrate that the proposed framework delivers reliable grasping performance for most daily-life objects, while highlighting the challenges of handling soft or low-profile items. These challenges can be addressed through

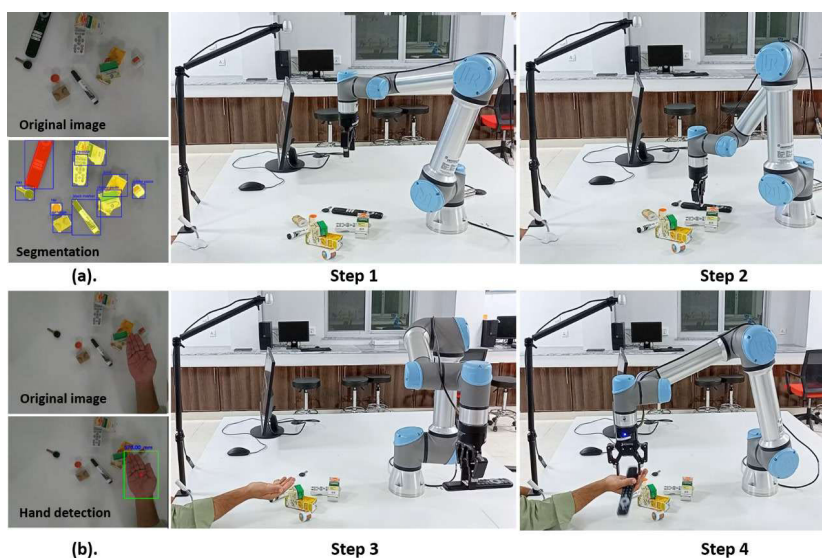


FIGURE 13. Voice command “Pick up the TV remote and the Sequence of robot operation to grasp the objects and handing over to the user while detecting and tracking the hand. a). target object detection and grasping action, and b). robust handover to the user with hand detection and tracking.

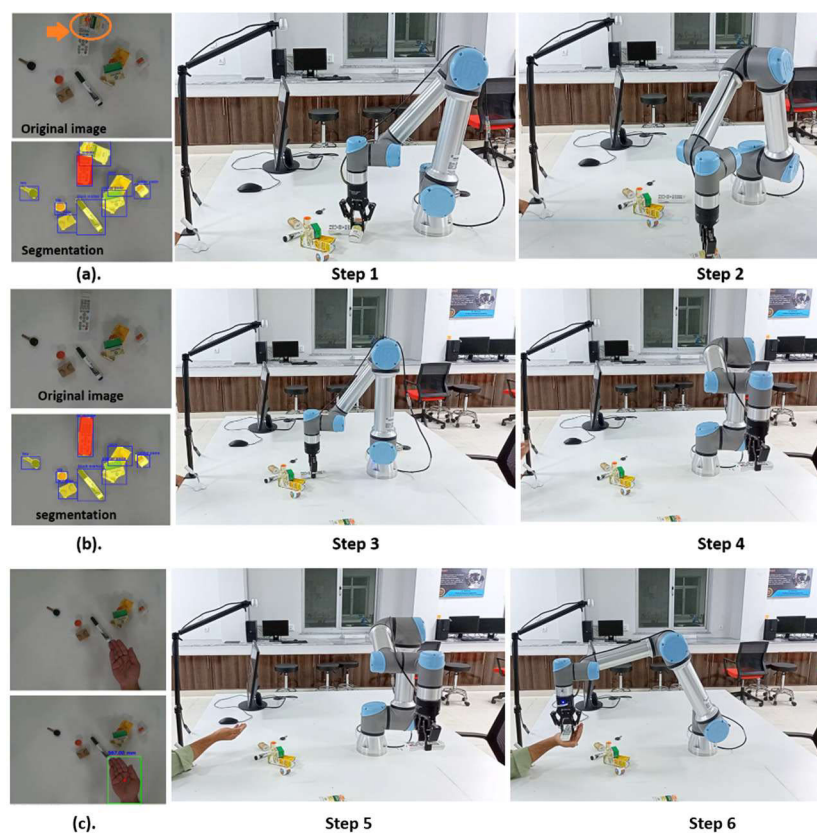


FIGURE 14. Voice command “Pick up the AC remote” and the Sequence of robot operation to grasp the objects and handing over to the user while detecting and tracking the hand. a). Occlusion detection, and occlusion removal robotic action. b). target object detection and grasping action, and c). robust handover to the user with hand detection and tracking.

adaptive gripper with different grasping modes, such as pinch grasping for deformable objects (e.g. Bandages), scoop based

grasping techniques for low profile objects (e.g., keys, and taps), and encompassing grasping for round or heavier objects

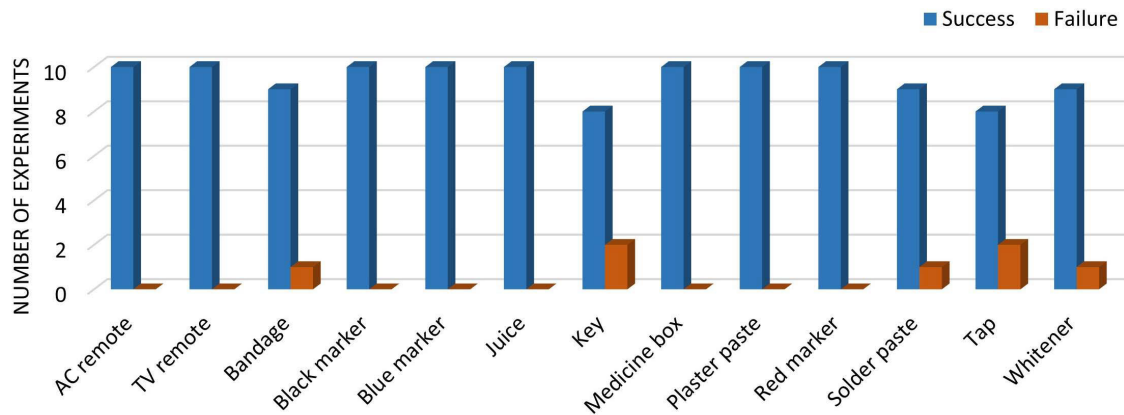


FIGURE 15. Evaluation of Grasp success and failure rates for thirteen everyday objects. Blue bars indicate the successful grasp, while orange bars indicate the failures. The results show how the object geometry and texture affect the grasp success rate.

TABLE 8. Quantitative evaluation of voice-to-grasp execution latency in the proposed assistive robotic system, demonstrating the system's responsiveness and practicality for assistive use.

S. No.	Voice Command	Speech Duration (s)	Recognition Latency (s)	Total Command Time (s)	Robotic Action		Total Robot Action (s)	Overall Time (s)
					Robot Grasping (s)	Handing Over (s)		
1	Pick up juice box	3.21	3.02	6.23	30.23	5.52	35.75	41.98
2	Pick up juice box	3.21	3.69	6.9	32.11	5.92	38.03	44.93
3	Grab the AC remote	3.6	1.65	5.25	31.22	6.05	37.27	42.52
4	Handover the bandage	3.27	2.06	5.33	30.91	6.25	37.16	42.49
5	Fetch the medicine box	2.98	2.48	5.46	31.52	6.54	38.06	43.52
6	Bring red marker	3.3	1.7	5.1	30.56	4.88	35.44	40.44
7	Robot give me solder paste	3.51	2.46	5.97	29.69	5.26	34.95	40.92
8	Pick up the whitener	3.49	2.1	5.59	32.06	6.6	38.66	44.25
9	Give me juice box	3.6	1.87	5.47	33.27	6.59	39.86	45.33
10	Hold the black marker	3.06	2.6	5.66	30.93	4.99	35.92	41.58
11	Take the tap	2.61	2.08	4.69	31.48	5.73	37.21	41.9
12	Get blue marker	3.78	2.36	6.14	30.15	5.34	35.49	41.63
13	Lift up the juice pack	3.0	1.81	4.81	32.84	6.41	39.25	44.06
14	Fetch the medicine box	2.98	2.48	5.46	31.91	5.87	37.78	43.24
15	Bring red marker	3.3	1.7	5.0	30.42	5.11	35.53	40.53
16	Robot give me solder paste	3.51	2.46	5.97	30.09	6.32	39.41	45.38
17	Take the tap	2.61	2.08	4.69	29.87	5.46	35.33	40.02
18	Lift up the juice pack	3.0	1.81	4.81	31.66	6.08	37.74	42.55
19	Pick up medical tap	3.11	3.29	6.4	30.75	5.67	36.42	42.82
20	Grab the TV remote	3.2	1.65	4.85	32.39	6.12	38.51	43.36
Average	—	3.22±0.31	2.27±0.56	5.48±0.63	31.35±1.01	5.84±0.54	37.19±1.43	42.67±1.47

(e.g., balls, apples, or similarly shaped items), where parallel grippers may fail due to weight or gravity effects.

VI. FAILURE CASES AND FUTURE WORK

Although the proposed assistive grasping system demonstrated high accuracy and stable low-latency performance, few challenging scenarios were identified during experimentation. The grasp failures (Figure 15) occurred primarily for soft (bandage) and low-profile objects (tap, key), where the

Robotiq gripper occasionally failed to achieve a stable hold due to insufficient contact surface. Additional failures were observed when the target object was tightly packed or fully occluded by surrounding items, leading to either unsafe grasp poses or the system reporting no detectable object within the workspace. Furthermore, certain grasp attempts were aborted when objects were positioned near the workspace boundary, as constrained by the predefined safety limits to prevent overextension of the robotic arm.

Future work will focus on developing a shape-aware, context-aware, language-guided target selection and occlusion-removal algorithm capable of handling densely cluttered and fully occluded environments. Especially, the introduction of nonlinear pushing trajectories which requires much fewer iterations as compared to linear pushing trajectories, thereby improving efficiency in occlusion and clutter removal.

For soft object grasping various soft grippers [69], [70] were designed and developed and for low profile objects, adaptive grippers [71] designs are suitable for grasping low profile object by different grasping types like pinch, or scoop grasp. The proposed system is planned to be evaluated with visually impaired participants in collaboration with rehabilitation institutes to obtain realistic user feedback and assess the system's practical usability, safety, and effectiveness in real-world assistive scenarios, after giving rigorous training sessions about the system usability and safety.

The current Google Voice Recognition API relies on an active internet connection and may experience additional delays under unstable network conditions. We also acknowledge that offline alternatives, such as VOSK or Whisper, could be integrated in future versions of the system. These offline models remove network dependency, enhance privacy, and can maintain real-time performance depending on the available hardware. In contrast, cloud-based services like Google Voice API support over 120 languages and provide advanced features such as real-time streaming recognition, automatic punctuation, and multi-speaker identification. Therefore, a hybrid approach could be a feasible solution that combines both cloud based and local models, that automatically switch between online and local offline models to ensure consistent operation.

VII. CONCLUSION

This research presents a complete end-to-end assistive grasping framework combining speech-based command recognition, intelligent text classification, and YOLOv11s-driven visual perception for robotic manipulation. The integration of multimodal modules enables sighted participants to control a robotic manipulator using natural spoken commands with high accuracy and safety, and it can be extended to the visually impaired user with proper training and guidance. Quantitative evaluations demonstrated superior segmentation and recognition performance, achieving an mAP_{50–95} of 0.933 for detection, 0.882 for segmentation, and 97.93% text classification accuracy. The overall system achieved an average grasp success rate of 94.7% and maintained near real-time operation with a 17.9 FPS perception rate and 4.2 ms inference latency per frame. Safety mechanisms, including workspace limits and force-torque feedback, ensured secure operation without unintended collisions. Future work will focus on improving grasp robustness in cluttered and occluded environments through adaptive grasp planning, multi-view perception, and context-aware language models. The proposed system establishes a foundation for intelligent

TABLE 9. UR5e universal robot specifications.

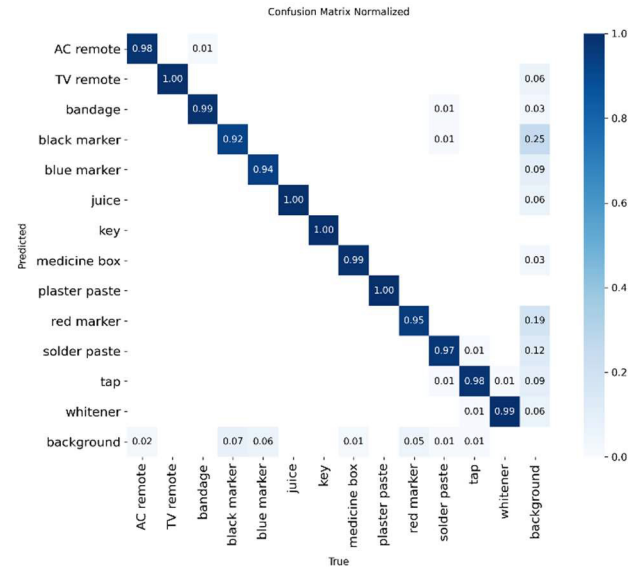
Parameters	Description
Robot type	UR5e
Robot weight	20.7 kg
Maximum payload	5 kg
Reach	850 mm
Joint ranges	Unlimited rotation of tool flange, $\pm 360^\circ$ for all joints
Speed	Joints: Max 180 °/s, Tool: Approx. 1 m/s.
System update Frequency	500 Hz
Force torque sensor accuracy	4N
Degrees of freedom	6 DOF
Tool I/O power supply and voltage	12 V/24 V 1.5 A
Communication	TCP/IP 1000 Mbit: IEEE 802.3ab, 1000BASE-T Ethernet socket, MODBUS TCP & EtherNet/IP Adapter, Profinet
Power consumption (average)	570 W
Materials	Aluminum, PC/ASA plastic

assistive robots that can perform safe, reliable, and intuitive object manipulation in everyday living environments.

APPENDIX

Universal Robot (UR5e) important specification are listed in the Table 9, while rest of the detailed specification and parameters can be seen in the web-site "www.universalrobots.com/manuals/EN/HTML/SW 5_19/Content/prod-usrman/complianceUR5e/H_g5_sections/appendix_g5/tech_spec_sheet.htm"

CONFUSION MATRIX



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MUHAMMAD ARSALAN received the B.Sc. and M.Sc. degrees in mechanical and mechatronics engineering from the University of Engineering and Technology, Peshawar, Pakistan, in 2016 and 2020, respectively. He is currently pursuing the Ph.D. degree with the Robotics and Automation Laboratory, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Pakistan. His research interests include robot vision, robotic manipulation, grasping in cluttered environments, vision-language models (VLMs), and agricultural robotics.



ABID IMRAN received the B.S. degree in mechatronics and control engineering from the University of Engineering and Technology (UET), Lahore, Pakistan, in 2011, and the integrated M.S. and Ph.D. degree in electronic systems engineering from Hanyang University, South Korea, in 2019. From March to August 2019, he was a Postdoctoral Fellow with the Research Institute of Engineering and Technology, Hanyang University. He is currently an Associate Professor with the

Faculty of Mechanical Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI), Pakistan. His research interests include human–robot interaction, motion optimization, gripper/hand design, and artificial intelligence.



KHURRAM KHAN received the B.S. degree in computer engineering from COMSATS University Islamabad, Pakistan, in 2007, and the M.S. and Ph.D. degrees in electronic and communication engineering from Hanyang University, South Korea, in 2010 and 2019, respectively. He is currently with the Faculty of Computer Science and Engineering, Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI), Pakistan. His research interests include computer

vision, pattern recognition, and machine/deep learning.



ARSALAN ARIF was born in 1984. He received the B.S. degree in electrical engineering from the University of Engineering and Technology (UET), Peshawar, Pakistan, in 2006, and the M.S. and Ph.D. degrees in electronics system engineering from the Department of Electronics System Engineering, Hanyang University, Ansan, South Korea, in 2010 and 2013, respectively. He is currently an Assistant Professor with the Faculty of Mechanical Engineering, Ghulam Ishaq Khan Institute of

Engineering Sciences and Technology (GIKI), Pakistan. His research interest includes design and control of electrical machines.



MUDASSIR RAZA SIDDIQI was born in 1991. He received the bachelor's degree in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2012, the M.S. degree in electrical engineering from the Energy Conversion System Laboratory, Hanyang University, South Korea, in 2017, and the Ph.D. degree from the Department of Electrical Engineering, Incheon National University, South Korea, in 2022. From 2017 to 2018, he was a

Researcher with Nex-M Protohouse, Ansan, South Korea. He was a Research Professor of electrical engineering with the Department of Electrical Engineering, Incheon National University, until April 2024. He has been a Postdoctoral Researcher with the Division of Electric Power and Energy Systems, KTH Royal Institute of Technology, Sweden, since May 2024. His research interest includes the design and control of electric motors.

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